Method of Means and Medians

When sinks are irregular, greedy merge adapts better than a pure geometric H

The idea

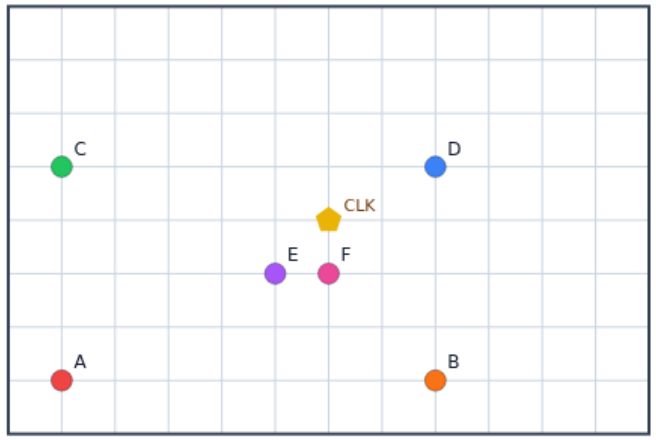
- Treat each sink as a fragment
- Repeatedly merge the closest pair of fragment roots at their midpoint

Means & medians

MMM TREE

Means & medians

Step 1 / 5 · idea · module02-03-mmm-tree



Explanation

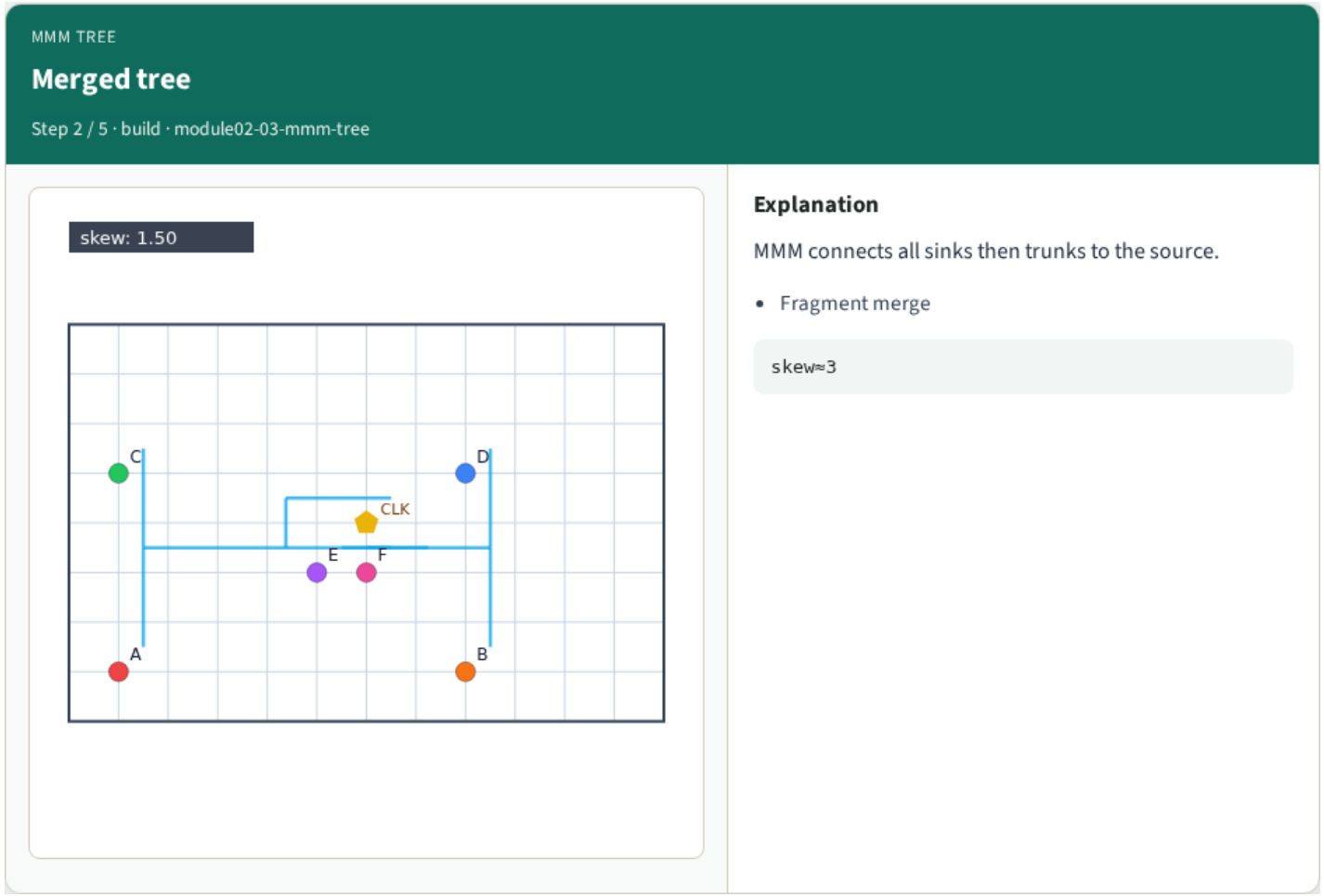
Repeatedly merge the closest fragment roots at their midpoint.

- Greedy merge
- Steiner at merge

```
Build MMM
```

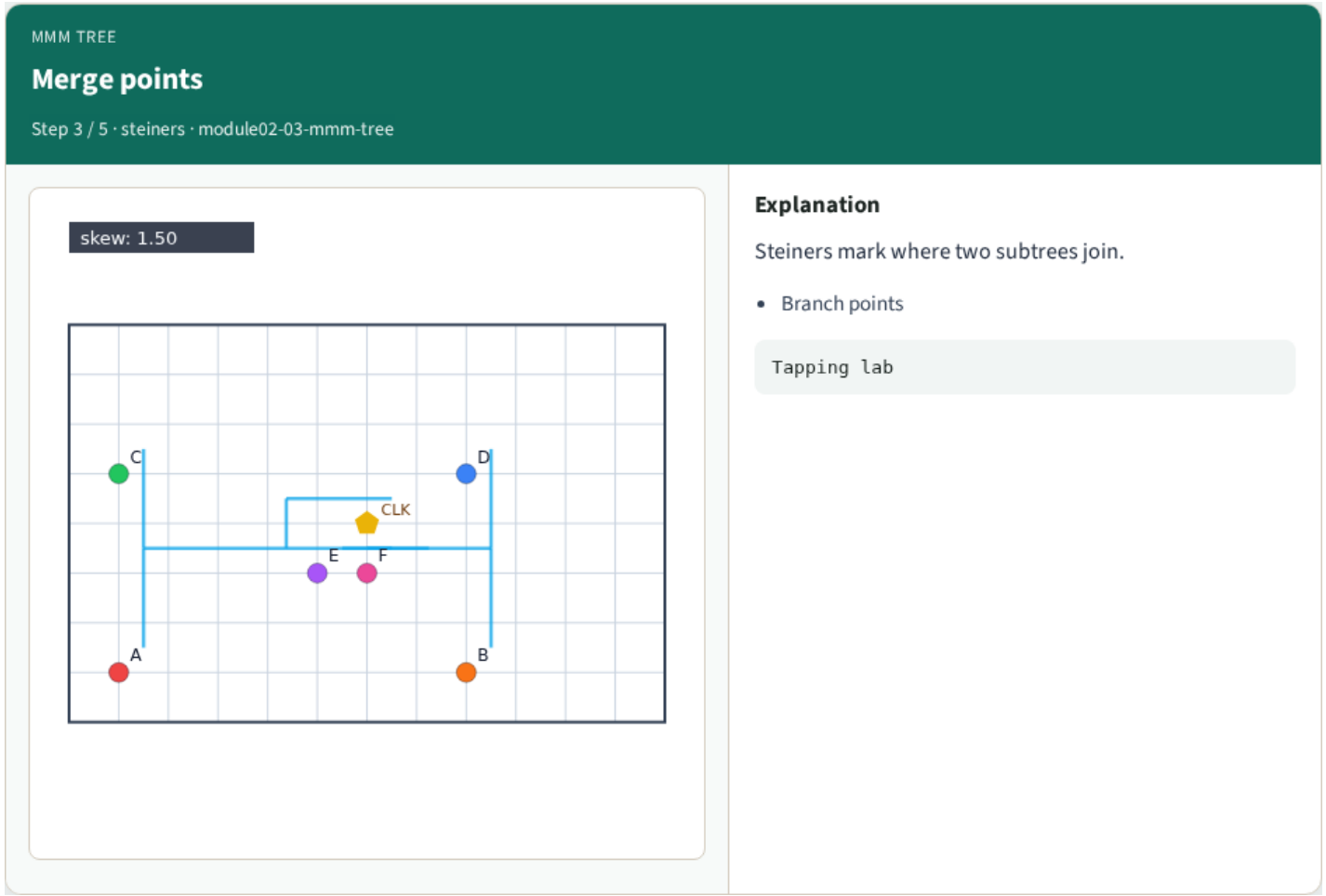
Means & medians

Merged tree



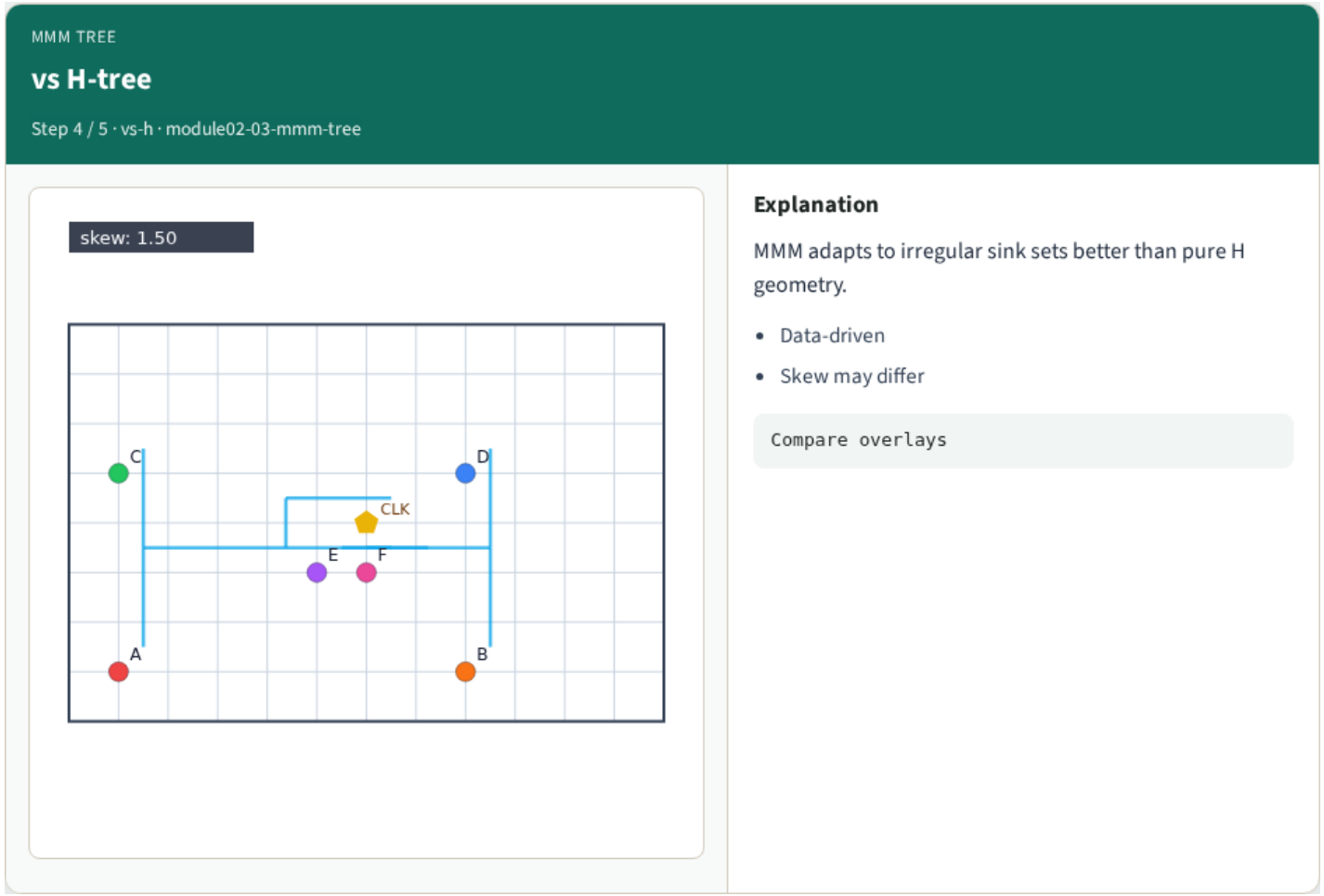
Merged tree

Merge points



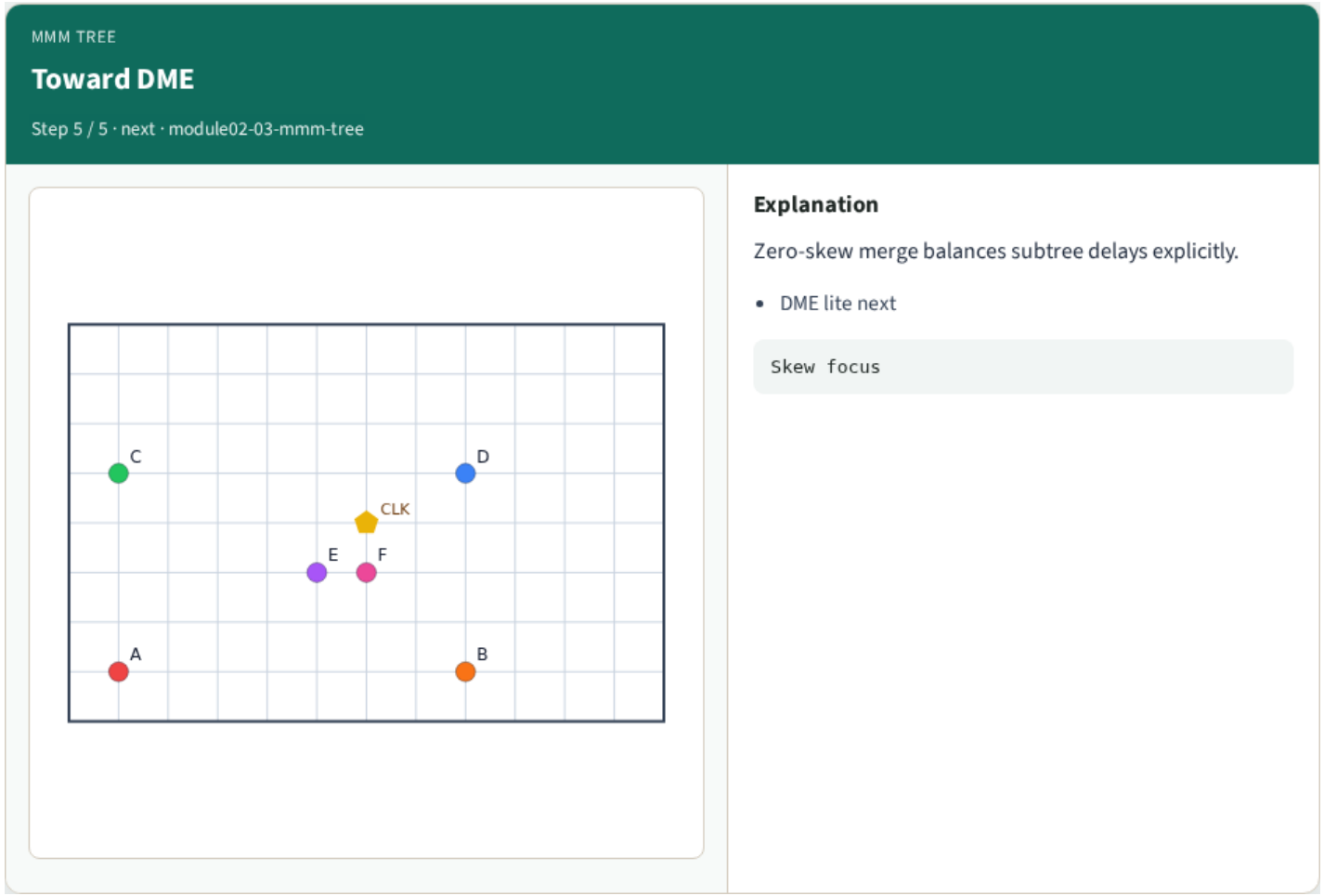
Merge points

vs H-tree



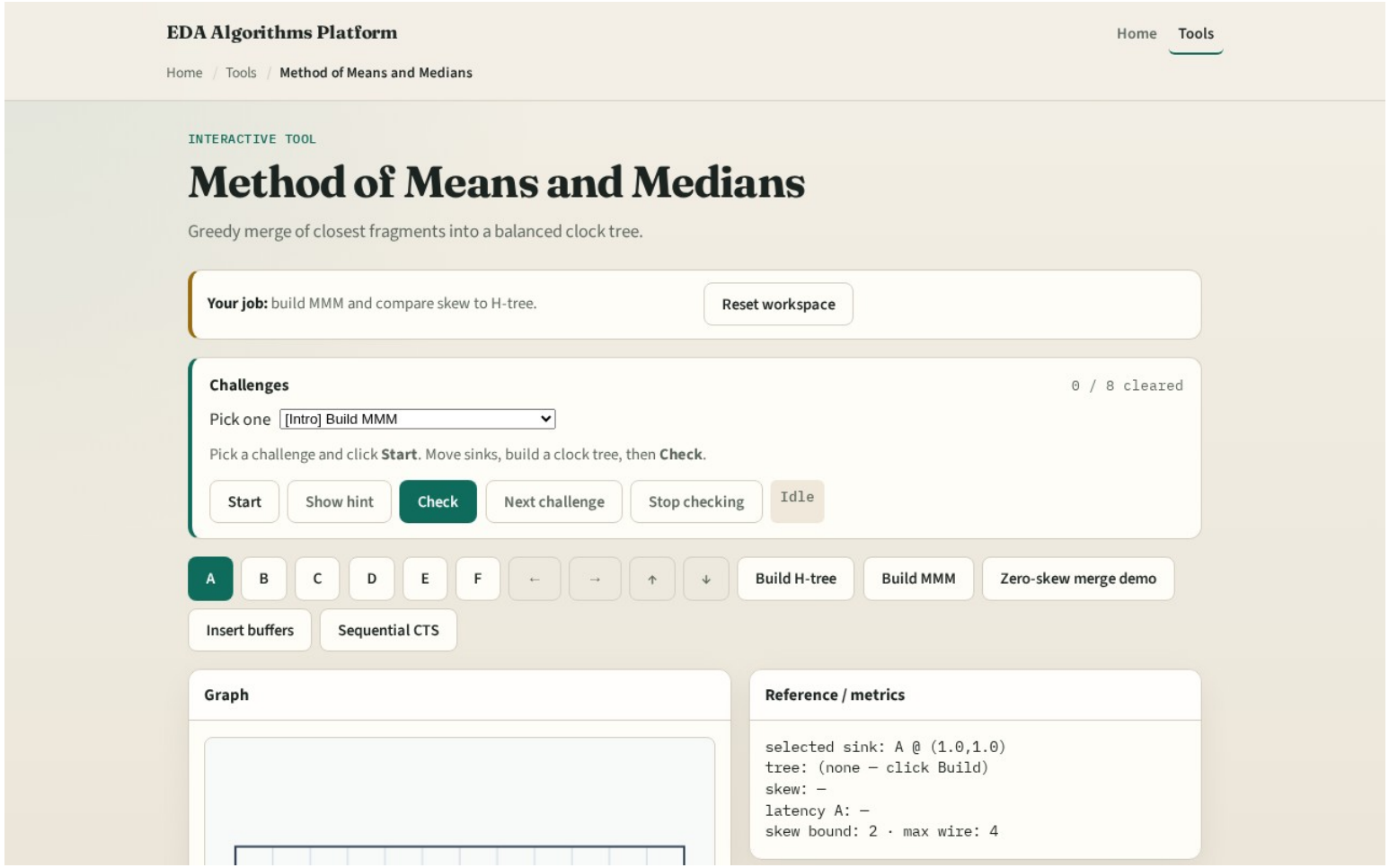
vs H-tree

Toward DME



Toward DME

Browser lab track



Browser lab starter

Implement track

- Call `mmm_tree(source, sinks)` in solvers
- Assert every sink endpoint appears in the edge set

Pitfalls

- Merging by Euclidean when latency is Manhattan
- Forgetting to connect the final fragment to the source
- Dropping steiner coordinates when buffering

Your turn

- Next: zero-skew merge (DME lite)